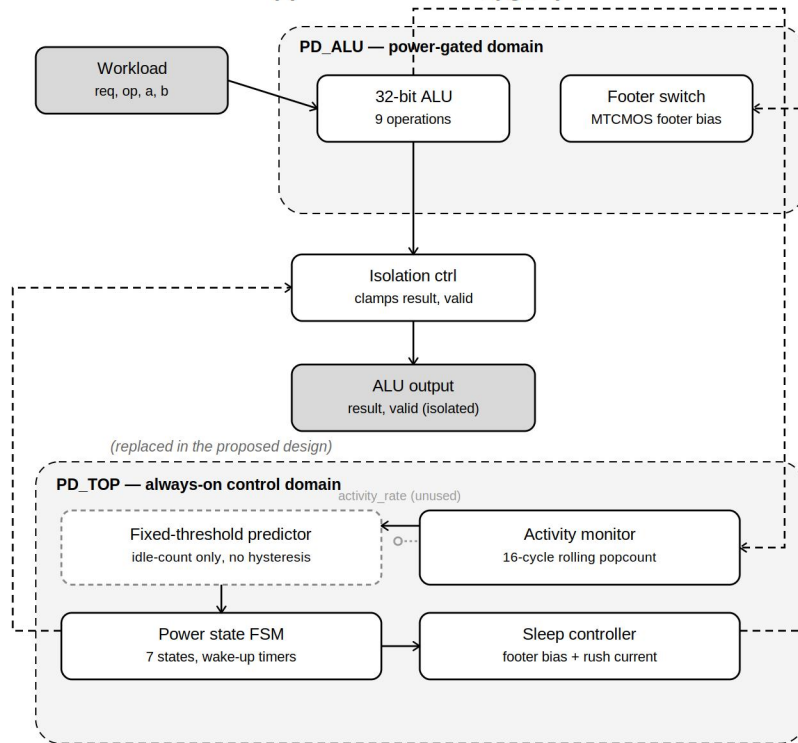


Baseline vs. Proposed Controller Architecture — Single-Module Substitution

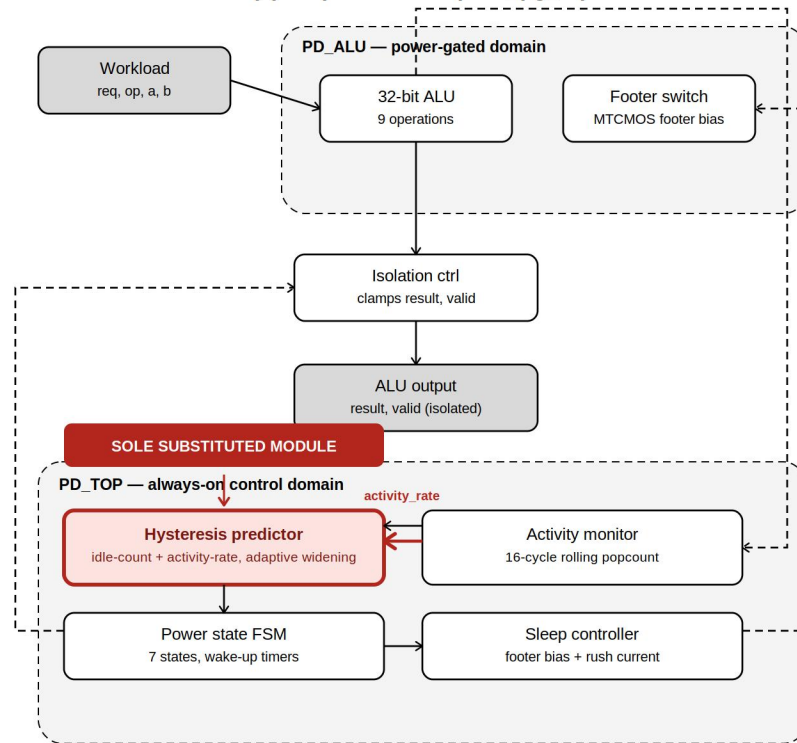
Every block is bit-for-bit identical between the two designs (same RTL, same UPF power intent) except the sleep-entry predictor.

(a) Baseline — static_pg_top



Post-route: 1014 cells · 361.50 μm^2

(b) Proposed — adaptive_pg_top



Post-route: 1118 cells · 410.61 μm^2 (+13.6% area, +104 cells)

 Shared / unmodified module (identical RTL in both designs)

 Sole substituted module — the paper's one experimental variable

Everything downstream of the predictor (FSM, sleep controller, isolation, ALU) receives the same sleep_req encoding and behaves identically in both designs.